TEMPERATURE CONTROLLER FAN

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**TEMPERATURE CONTROLLER FAN"** has been carried out by the students **SUWATHI S, SRI VARDHINI T, SUDHARSHINI R,** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

This project presents a temperature-controlled fan system designed to automatically regulate airflow based on ambient temperature variations. The system uses a temperature sensor to continuously monitor the surrounding environment and a microcontroller to process the sensor data. When the measured temperature exceeds a predefined threshold, the controller activates the fan and adjusts its speed proportionally to the temperature level, ensuring efficient cooling. This automation reduces energy consumption, enhances user comfort, and eliminates the need for manual intervention. The design is simple, cost-effective, and suitable for applications such as home appliances, electronic equipment cooling, and smart environmental control systems.This project presents a temperature-controlled fan system designed using an **Arduino Uno**, **DHT11 temperature sensor**, **L298N motor driver**, and a **12V exhaust fan** to provide automatic cooling based on real-time environmental conditions. The DHT11 sensor continuously measures the ambient temperature and sends the data to the Arduino, which processes the readings and activates the fan when the temperature exceeds a predefined threshold. The L298N motor driver enables safe and efficient control of the high-power 12V fan using low-power signals from the Arduino.

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SIGNATURE

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LIST OF ABBREVIATIONS

AC	-	Alternating Current
DC	-	Direct Current
GND	-	Ground
RH	-	Relative Humidity
I2C	-	Inter-Integrated Circuit
IDE	-	Integrated Development Environment
MCU	-	Microcontroller Unit
LED	-	Light Emitting Diode
LCD	-	Liquid Crystal Display
DHT	-	Digital Humidity and Temperature Sensor
UNO	-	Arduino Uno Microcontroller Board
L298N	-	Dual H-Bridge Motor Driver Module
PWM	-	Pulse Width Modulation
RPM	-	Revolutions Per Minute
PCB	-	Printed Circuit Board
USB	-	Universal Serial Bus
mA	-	Milliampere
VCC	-	Voltage Common Collector
Hz	-	Hertz

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

This project focuses on developing a fan control mechanism that automatically adjusts its operation according to the surrounding temperature. Using a temperature sensor to continuously monitor the environment, the system processes the data through a control unit—typically a microcontroller—and activates or varies the speed of the fan as needed. This intelligent control not only enhances user comfort but also conserves energy by running the fan only when necessary. Temperature control plays a vital role in maintaining comfort, protecting electronic devices, and ensuring efficient ventilation in both residential and industrial environments. Traditional fans require manual operation and often run continuously, leading to unnecessary power consumption and inconsistent cooling. With the growing demand for automation and energy-efficient systems, microcontroller-based temperature-controlled fans have become an effective solution for intelligent environmental management.

This project presents a temperature controller fan system developed using an Arduino Uno**,** DHT11 temperature sensor**,** L298N motor driver**,** and a 12V exhaust fan**.** The DHT11 sensor continuously monitors the ambient temperature and sends the data to the Arduino, which processes the readings and automatically activates the fan when the temperature exceeds a predefined threshold. The entire circuit is assembled on a **b**readboard using jumper wires and powered through a 12V adapter connected via a female DC connector. A cardboard enclosure is used to simulate a small room or ventilation chamber, demonstrating the practical working of the system. By integrating low-cost electronic components with simple automation logic, this project showcases how embedded systems can be used to create smart, energy-efficient cooling solutions. It highlights the importance of automation in modern applications and demonstrates how microcontroller-based systems can improve convenience, reduce power consumption, and enhance environmental control. In essence, this project illustrates how basic analog principles can be leveraged to design smart, automated cooling systems that contribute to improved safety, performance, and sustainability in electronic applications.

1.2 BACKGROUND

The background of this project is to design and develop an automatic temperature-controlled fan system that can regulate its operation based on real-time changes in ambient temperature. The system aims to accurately sense temperature variations using a suitable sensor and adjust the fan's speed accordingly to maintain a comfortable and stable environment. By automating the cooling process, the project seeks to reduce unnecessary energy consumption, improve user convenience, and enhance the overall efficiency of temperature management. Additionally, the objective includes creating a simple, reliable, and cost-effective solution that can be applied in homes, offices, electronic equipment cooling, and various smart automation applications.

The main objective of this project is to design and implement an automatic temperature-controlled fan system using Arduino Uno**, DHT**11 sensor**,** L298N motor driver**,** and a 12V exhaust fan to provide efficient and intelligent cooling based on real-time temperature conditions. The system aims to continuously monitor ambient temperature and activate the fan automatically when the temperature exceeds a predefined threshold. The cardboard enclosure is used to simulate a small room or ventilation chamber for practical demonstration. Overall, the project seeks to create a low-cost, energy-efficient, and user-friendly temperature regulation system using basic embedded hardware components. By using a temperature sensor and a microcontroller, the project focuses on improving user comfort, reducing energy consumption, and enhancing efficiency in cooling systems. This automatic control mechanism helps maintain a comfortable indoor environment while promoting energy conservation and demonstrating the practical application of embedded systems and sensor-based automation.

CHAPTER 2

LITERATURE SURVEY

Zhao. X, and Liu. H (2025) introduced an adaptive thermal regulation module using a hybrid analog–digital architecture composed of NTC thermistors, PWM-controlled cooling fans, and microcontroller-based feedback control loops. Their design offered high precision and dynamic control of fan speed to minimize heat stress in compact electronic devices. However, the requirement of firmware programming, ADC calibration, and frequent software tuning increased system complexity and development cost. Moreover, the hybrid structure made troubleshooting more difficult for low-level users and limited the design's applicability in low-budget or academic environments.

Singh .M, and Prakash. D (2024) developed an analog comparator-based thermal shutdown system for power supply units utilizing an LM358 op-amp and resistive sensing network. The primary objective was to minimize overheating in transformer-based circuits. Although the system demonstrated quick response time and reliable threshold switching, it lacked multi-stage cooling functionalities and was highly sensitive to supply voltage fluctuations. Their work emphasized the importance of stable reference voltage design, a limitation that this project addresses through calibrated potentiometer adjustment.

Verma. S and Bhatia. P (2024) evaluated thermistor linearization techniques for use in precision temperature-controlled environments. They proposed various resistor network configurations to improve the sensitivity range of NTC thermistors between 25°C and 45°C. While increasing accuracy, the design required more components and careful tolerance selection. Their findings, however, highlight the enduring relevance of thermistors in low-cost thermal control applications.

Nakamura. H and Ito. Y (2023) investigated a transistor-based fan driver using high-gain NPN power transistors for cooling microelectronic components. Their system integrated thermal feedback loops but exhibited risks of thermal runaway when operating under high ambient temperatures. The study underscores the necessity of proper diode protection and reliable heat dissipation mechanisms—features incorporated in the present project through the use of a BC140 transistor and flyback diode.

Das. R, and T. Roy (2023) designed a compact analog cooling circuit using an LM741 op-amp for comparator-based switching, aimed at temperature-controlled laboratory instruments. Their work demonstrated that analog systems remain highly effective for environments requiring low latency and minimal noise. However, they noted limitations such as reference drift and varying switching thresholds under different power supply conditions. This reinforces the importance of simple and stable analog designs like the one implemented in this project.

Hernandez. R and M. Flores (2022) developed a fan speed control system using thermistor sensing combined with MOSFET-based switching. The focus was on optimizing energy consumption in household appliances. Although the system provided efficient switching, the MOSFET's gate sensitivity necessitated strict noise filtering and precise reference voltage levels. Their findings highlight the practicality of transistor-driven cooling systems but also demonstrate why bipolar linear drivers are preferred in simpler academic circuits.

Chen. L and Zhang. J (2021) proposed a low-cost analog cooling system for embedded microprocessors using an op-amp comparator and temperature-dependent voltage divider. Their approach was centered around affordability and ease of implementation, making it suitable for student projects and small-scale applications. Yet, their design lacked protective mechanisms against inductive loads, limiting its adaptability for larger DC fans. This shortcoming is addressed in the present system through the inclusion of a diode-based protection circuit.

Peters. K and Johansson. S (2020) focused on long-term thermal reliability in low-power analog circuits using continuous temperature feedback loops. Their findings indicated that analog automation is significantly more power-efficient than digital approaches in small-scale applications. However, generalizability was limited because the system relied heavily on precision op-amps not commonly available in low-resource academic settings

Morgan. J and Harris. B (2019) examined the viability of analog temperature control systems in industrial environments. They concluded that such systems are robust and fast-responding but require special care in calibration to maintain accuracy over extended periods. Their work highlights the trade-off between simplicity and long-term drift—an important consideration in the current project's design.

Patel. S and Mehra. K (2018) presented one of the earlier studies involving thermistor-based fan regulation for small communication devices. Their analog prototype demonstrated the fundamental principles of comparator-based switching. However, the lack of a transistor driver restricted fan current, causing inefficiencies under load. This limitation directly influenced the development of the present project, which employs a high-current BC140 transistor to overcome such drawbacks.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1. EXISTING METHOD

In the traditional or existing system, fan speed control is predominantly manual and does not depend on the ambient room temperature. Conventional ceiling fans, table fans, and exhaust fans operate at fixed speed levels that are controlled by a mechanical regulator, electronic dimmer, or wall-mounted switch. Users must manually adjust the speed according to their comfort level, which often leads to inefficient operation and unnecessary energy consumption. These systems lack any form of intelligence or automation and depend entirely on human intervention for speed regulation. Most conventional fans use either step-type regulators or electronic regulators. Step regulators provide discrete speed levels such as low, medium, and high by changing the resistance in the circuit. This method causes power loss in the form of heat and reduces energy efficiency. Electronic regulators improve efficiency slightly by using triacs or capacitive control, but they still require manual adjustment and do not respond dynamically to changes in temperature. As a result, the fan may continue to operate at a high speed even when the room temperature has dropped, wasting electrical energy. In some modern systems, fans are integrated with remote controls or smart switches. While these provide convenience in terms of remote operation, they still rely on user input rather than environmental conditions. The user must observe the temperature and manually increase or decrease the fan speed. This approach is not only inconvenient but also unreliable, especially during sleep or when the user is away from the room. These systems do not ensure optimal thermal comfort or energy efficiency.

In industrial and commercial environments, centralized air-conditioning or ventilation systems are sometimes used to regulate temperature. Although these systems may include temperature sensors, they are expensive, complex, and consume significant power. They are not suitable for small residential applications due to high installation and maintenance costs. Moreover, traditional fan systems in such environments still operate independently of real-time temperature data. Another limitation of existing systems is the absence of feedback mechanisms. Conventional fans do not monitor or display room temperature, making it difficult for users to make informed decisions regarding speed control. This often results in discomfort due to overcooling or undercooling. Additionally, continuous operation at higher speeds increases wear and tear on fan components, reducing their lifespan and increasing maintenance requirements. In summary, the existing fan control systems are largely manual, inefficient, and non-adaptive. They lack automatic temperature sensing, real-time feedback, and intelligent control mechanisms. These drawbacks highlight the need for an automated temperature-based fan control system that can adjust fan speed dynamically based on room temperature, improve energy efficiency, enhance user comfort, and reduce manual effort. In the context of academic and prototype environments, complex digital cooling systems are often impractical due to their high dependency on programming, debugging, and expensive sensors. On the other hand, manually operated systems fail to deliver the precision and automation required for modern electronic applications. These limitations highlight the need for a system that is automatic**,** reliable**,** low-cost, and easy to integrate into existing electronics.

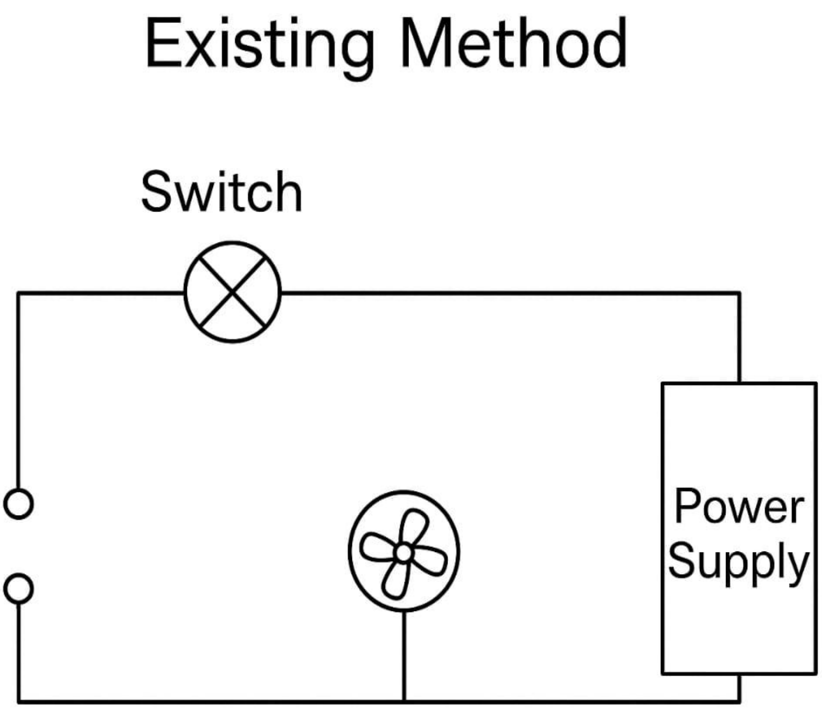


Figure 2.1 Existing Circuit Diagram

3.2. PROPOSED METHOD

The proposed method focuses on designing and implementing an automatic temperature-controlled fan system that operates without manual intervention and without the use of a display unit. The primary goal of this system is to automatically regulate the speed of a fan based on real-time ambient temperature, thereby improving energy efficiency, user comfort, and system reliability. Unlike conventional fan systems that require continuous user adjustment, the proposed method uses a temperature sensor, a microcontroller, and a motor control circuit to achieve intelligent control. The system begins with the sensing of room temperature using a temperature sensor such as the LM35 or DHT11. This sensor is placed in the environment where temperature monitoring is required. The sensor continuously measures the surrounding temperature and converts it into an electrical signal proportional to the temperature variation. This signal represents the real-time thermal condition of the room and forms the input for the control system. The output of the temperature sensor is connected to the analog input pin of a microcontroller, such as the Arduino Uno. The microcontroller acts as the core processing unit of the system. It continuously reads the sensor output and converts the analog signal into digital data using its built-in analog-to-digital converter (ADC). The obtained digital temperature value is then compared with predefined temperature threshold levels stored in the microcontroller's program memory.

Based on the comparison results, the microcontroller determines the appropriate fan speed required to maintain comfortable room conditions. The control logic is designed such that when the room temperature is low, the fan either operates at a very low speed or remains switched off. As the temperature rises beyond a predefined lower threshold, the microcontroller gradually increases the fan speed. If the temperature reaches higher levels, the fan is driven at medium or maximum speed. This stepwise or proportional control ensures smooth operation and avoids sudden speed fluctuations. Fan speed control is achieved using Pulse Width Modulation (PWM) technique. The microcontroller generates PWM signals at one of its digital output pins. The motor driver acts as an interface between the low-power microcontroller and the high-power fan motor. By varying the duty cycle of the PWM signal, the effective voltage supplied to the fan motor is controlled, thereby regulating its speed.

A DC fan is used in the proposed method for ease of control and safety. The fan receives power from an external power supply suitable for its voltage and current ratings. A common ground is maintained between the microcontroller and the motor driver circuit to ensure proper signal reference. Protective components such as base resistors and flyback diodes are used to prevent damage due to voltage spikes and excessive current flow. The absence of an LCD display in this system simplifies the overall design and reduces cost, power consumption, and hardware complexity. Instead of displaying temperature values, the system operates entirely in an automated manner. Users do not need to monitor or adjust any parameters manually. The fan speed automatically adapts to environmental conditions, making the system suitable for continuous and unattended operation. This proposed method offers significant advantages over traditional fan control systems. It minimizes energy wastage by operating the fan only at the required speed. It also improves user comfort by maintaining a consistent cooling effect.

Since the fan does not run unnecessarily at high speeds, mechanical stress and wear are reduced, thereby increasing the lifespan of the fan. Additionally, the system is compact, low-cost, and easy to install, making it ideal for residential and small commercial applications. The system can further be enhanced in the future by incorporating wireless communication modules, mobile app control, or integration with smart home automation platforms. However, even in its basic form without a display unit, the proposed method provides an efficient, reliable, and intelligent solution for temperature-based fan speed control.

3.2.1. BLOCK DIAGRAM

The system consists of an Arduino Uno a s the main controller. A DHT11 sensor is connected to the Arduino to measure the ambient temperature. A 12V exhaust fan is driven through an L298N motor driver module, which acts as an interface between the low-voltage Arduino control signals and the high-voltage fan. The entire circuit is powered using a 12V adapter connected through a female DC connector. From this supply, power is given to the L298N driver and, through voltage regulation (either onboard or external), to the Arduino and other modules. All interconnections are made on a breadboard using jumper wires, and the setup is placed inside or attached to a cardboard enclosure representing a small room or chamber. The block diagram of the temperature controller fan system shows how each component interacts to measure temperature and operate the fan. The DHT11 sensor continuously senses the ambient temperature and sends the data to the Arduino Uno, which acts as the main controller.

The L298N motor driver is connected to the Arduino and controls the 12V exhaust fan. Since the fan requires higher voltage and current than the Arduino can supply, the L298N safely drives the fan using signals from the Arduino. The entire system is powered by a 12V adapter connected through a female DC connector, which supplies power to both the motor driver and the Arduino (through its onboard regulator). All wiring connections are made on a breadboard using jumper wires, and the setup is placed inside a cardboard enclosure to simulate a small room or chamber. This block diagram represents the flow of data (from sensor to Arduino to display) and the flow of power (from adapter to motor driver and fan), showing how the system works together to monitor temperature and operate the fan.

The block diagram of the temperature controller fan system illustrates the interaction between various components used to monitor and respond to temperature changes. At the core of the system is the Arduino Uno, which acts as the microcontroller and receives temperature data from the DHT11 sensor**.** This sensor continuously measures the ambient temperature and sends digital signals to the Arduino. The Arduino sends control signals to the **L298N** motor driver, which safely drives the 12V exhaust fan. The motor driver acts as an interface between the low-power control signals from the Arduino and the high-power requirements of the fan. The entire system is powered by a 12V adapter connected through a female DC connector, supplying power to both the motor driver and the Arduino. All components are interconnected using jumper wires on a breadboard, allowing easy prototyping and testing. The setup is housed within a cardboard enclosure, simulating a small room or chamber where temperature regulation is demonstrated

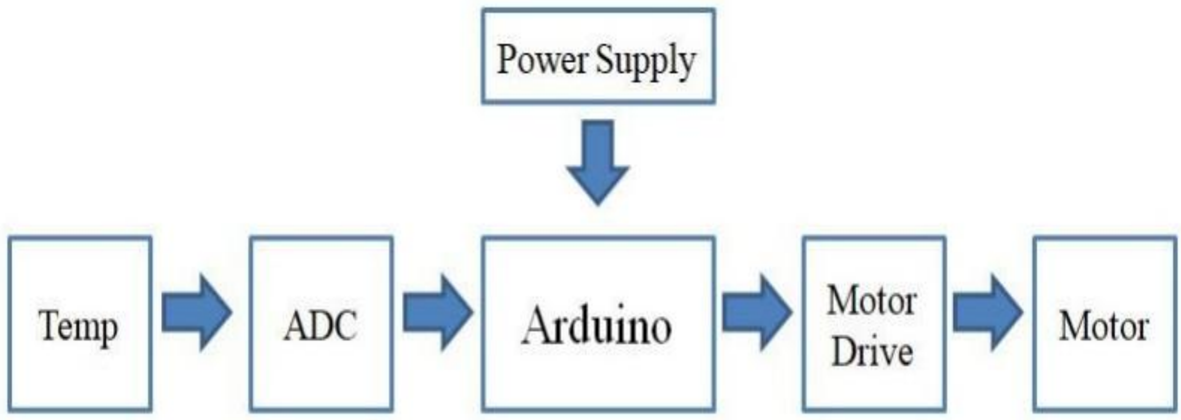


Figure 3.1 Block Diagram

3.2.2 CIRCUIT DIAGRAM

The circuit diagram of the automatic temperature-controlled fan project represents the electrical interconnection of all hardware components required to sense ambient temperature and regulate fan speed automatically. The design eliminates manual control and does not include any display unit, thereby simplifying the circuit and reducing power consumption and cost. The main elements of the circuit include a temperature sensor, a microcontroller, a motor driver stage, a DC fan, and a regulated power supply. The power supply section provides the necessary operating voltages to the entire circuit. A regulated 5V DC supply is used to power the microcontroller and the temperature sensor. This voltage may be derived from a 12V adapter using a voltage regulator or through the onboard voltage regulation of the microcontroller board. The DC fan operates at a higher voltage, typically 12V, which is supplied through an external adapter. It is essential that the ground terminal of the fan power supply is connected to the microcontroller ground to maintain a common reference point. This common grounding ensures proper signal transmission between the control and power stages of the circuit.

The temperature sensor, such as LM35 or DHT11, is responsible for continuously monitoring the surrounding room temperature. In the circuit, the sensor is connected directly to the microcontroller. The sensor's VCC pin is connected to the 5V supply, and the ground pin is connected to the common ground. The output pin of the sensor provides an analog voltage proportional to the measured temperature. This output is connected to one of the analog input pins of the microcontroller. As the temperature changes, the output voltage varies accordingly, enabling real-time temperature monitoring. The microcontroller serves as the central control unit of the system. It receives the temperature signal from the sensor through its analog input pin. The built-in analog-to-digital converter (ADC) converts the analog voltage into a digital value that represents the room temperature. The microcontroller continuously processes this data and compares it with predefined temperature threshold values stored in its program memory. Based on the comparison, the microcontroller decides the appropriate speed at which the fan should operate. To implement speed control, the microcontroller uses a Pulse Width Modulation (PWM) technique. A digital output pin capable of generating PWM signals is selected for this purpose. The duty cycle of the PWM signal is varied according to the temperature, allowing smooth and efficient control of the fan speed.

Since the microcontroller cannot directly drive the fan due to current and voltage limitations, a motor driver stage is used. This stage consists of a power transistor such as TIP122 or BC547, along with a base resistor. The PWM output from the microcontroller is connected to the base of the transistor through the resistor. The resistor limits the base current and protects both the transistor and the microcontroller. The collector of the transistor is connected to the negative terminal of the DC fan, while the emitter is connected to ground. When the microcontroller outputs a PWM signal, the transistor switches on and off rapidly, controlling the amount of current flowing through the fan. This switching action effectively regulates the fan speed according to the PWM duty cycle. A flyback diode is connected across the fan terminals to protect the transistor from voltage spikes generated by the inductive load of the fan. This diode ensures safe and reliable operation of the motor driver circuit.

The DC fan is the output device of the system and is responsible for providing cooling. The fan's positive terminal is connected to the external power supply, while the negative terminal is controlled through the transistor. As the temperature increases, the microcontroller increases the PWM duty cycle, allowing more current to flow through the fan, thereby increasing its speed. When the temperature decreases, the duty cycle is reduced, slowing down the fan or turning it off completely. In the complete circuit, all components work together to achieve automatic temperature-based fan control. The sensor continuously measures the room temperature and sends the data to the microcontroller. The microcontroller processes this information and generates a corresponding PWM signal. The motor driver amplifies this signal to control the fan speed. The absence of an LCD display makes the system fully automatic, requiring no user interaction.

The circuit design is simple, compact, and cost-effective. It reduces manual effort, conserves energy, and improves comfort. The elimination of a display unit further reduces hardware complexity and power usage. The modular design allows easy troubleshooting and future upgrades. The use of proper grounding, current-limiting resistors, and protective diodes enhances the reliability and safety of the circuit. The system is suitable for continuous operation and can be easily adapted for residential and small commercial environments.

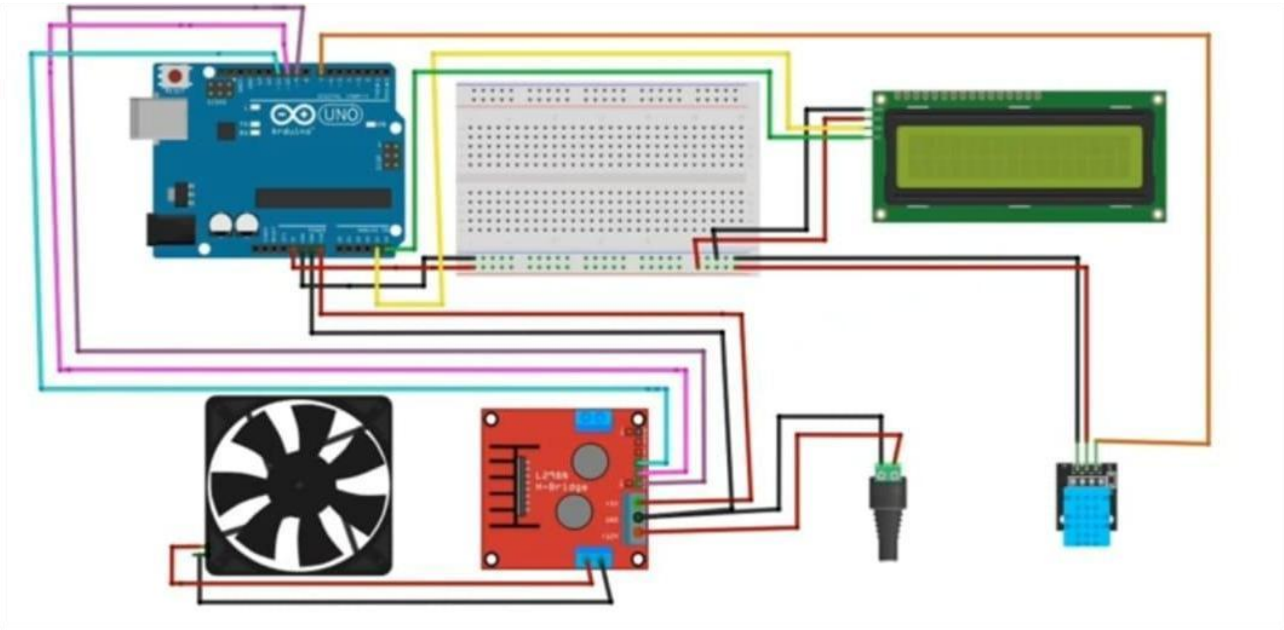
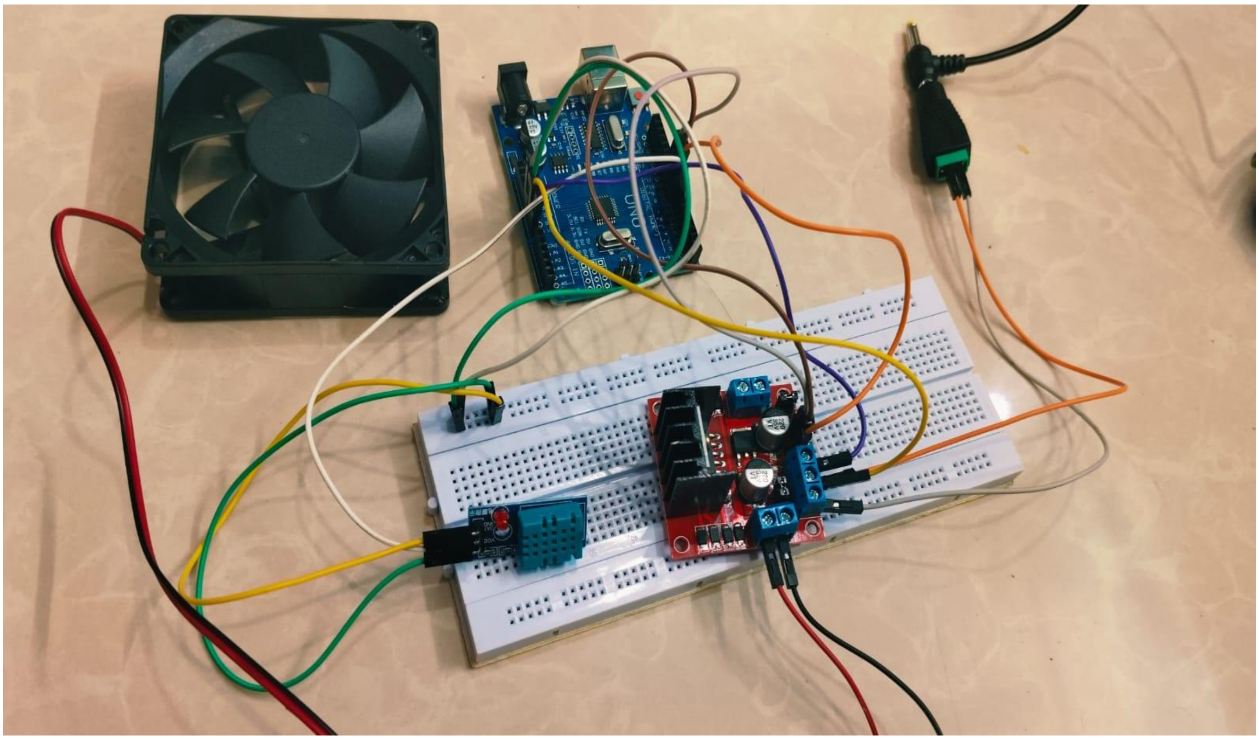


Figure 3.2 Circuit Diagram

The circuit diagram of the temperature controller fan system shows the interconnection of all components required for sensing temperature, processing data, and controlling the exhaust fan. The DHT11 sensor is connected to one of the digital pins of the Arduino Uno, with its VCC and GND pins connected to the 5V and ground rails on the breadboard. The L298N motor driver is connected to the Arduino through its input pins (IN1, IN2, and ENA), which receive control signals for operating the 12V exhaust fan**.** The fan is connected to the output terminals of the L298N, enabling it to run safely using the required voltage and current. A 12V adapter connected through a female DC connector supplies power to the L298N module, which in turn powers the fan, while the Arduino can be powered either from the same adapter or through the Arduino cable. All components are interconnected using jumper wires on a breadboard, ensuring proper distribution of power and signals. The entire circuit is mounted inside a cardboard enclosure, representing a small room or chamber where the temperature-controlled fan operation can be demonstrated effectively.

3.2.3. WORKING MODEL



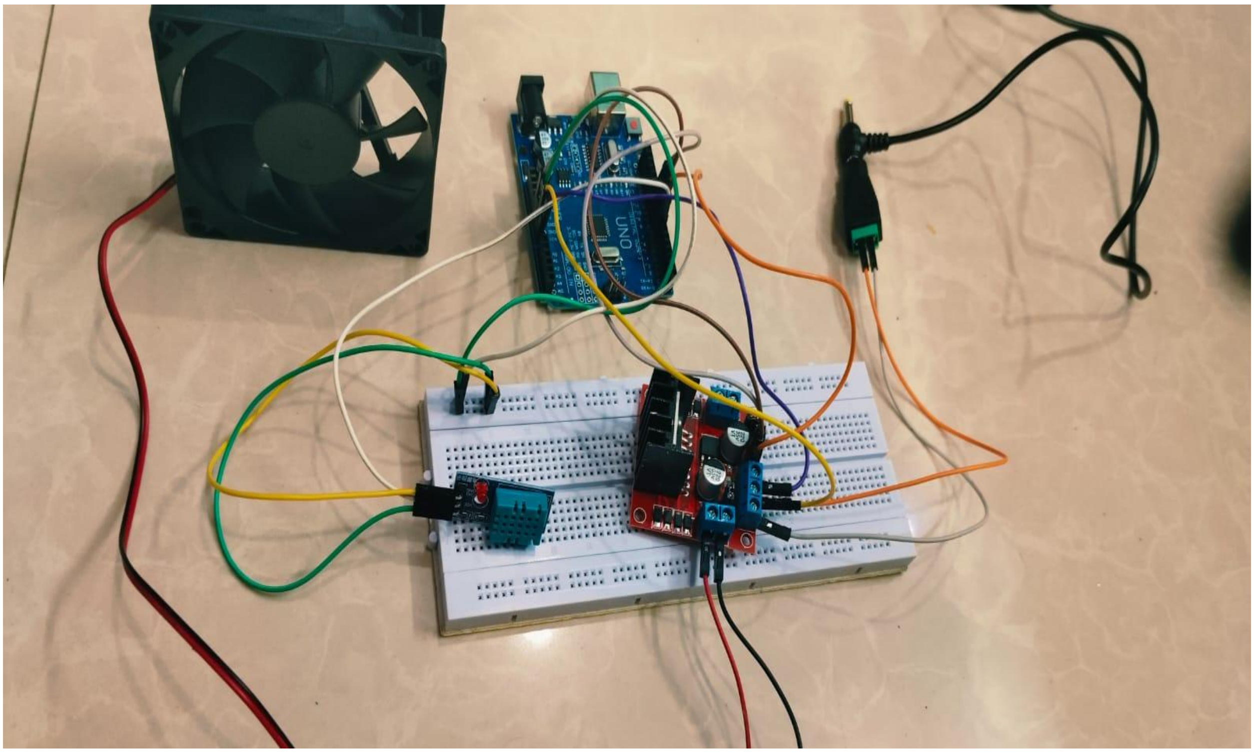


Figure: 3.3 Working Model

The working model of the automatic temperature-controlled fan consists of an Arduino Uno, a DHT11 temperature sensor, an L298N motor driver module, a DC fan, a breadboard, and an external power supply. The DHT11 sensor continuously measures the surrounding room temperature and sends the temperature data to the Arduino Uno in digital form. The Arduino acts as the central processing unit of the system, where the received temperature values are compared with predefined temperature limits programmed in the microcontroller. Based on the detected temperature, the Arduino generates appropriate control signals to regulate the fan speed.

These control signals are sent to the L298N motor driver module, which acts as an interface between the low-power Arduino and the high-power DC fan. Since the Arduino cannot directly drive the fan due to current limitations, the motor driver amplifies the control signals and supplies sufficient power to the fan. As the temperature increases, the Arduino increases the control signal intensity, causing the motor driver to rotate the fan at a higher speed. Similarly, when the temperature decreases, the control signal is reduced, and the fan speed slows down or stops completely. An external power adapter supplies the required voltage to the motor driver and fan, while all components share a common ground to ensure stable operation. Overall, the working model demonstrates an efficient and fully automatic system that adjusts fan speed according to room temperature without manual intervention, thereby improving comfort and conserving energy.

3.2.4. APPLICATIONS

The automatic temperature-controlled fan system has a wide range of applications in residential, commercial, and industrial environments due to its ability to regulate fan speed based on ambient temperature without manual intervention. One of the primary applications of this system is in residential homes, where it can be used in bedrooms, living rooms, and kitchens to maintain a comfortable indoor environment. By automatically adjusting the fan speed according to room temperature, the system enhances user comfort while reducing unnecessary power consumption, especially during night-time or when occupants are not actively adjusting the fan.

In office spaces and educational institutions, this system can be used to provide efficient cooling in classrooms, laboratories, and workspaces. Automatic fan control ensures a consistent airflow suitable for varying occupancy levels and environmental conditions. It reduces the dependency on users to manually change fan settings, thereby promoting energy-efficient practices in large shared spaces.

The system is also highly suitable for industrial applications, such as cooling of electronic equipment, control panels, and machinery. Many industrial devices generate heat during operation, and excessive temperatures can affect performance and lifespan. The temperature-controlled fan helps maintain optimal operating temperatures, preventing overheating and improving equipment reliability. In computer systems and server rooms, the automatic fan controller can be used to regulate cooling fans based on temperature variations. This ensures efficient thermal management, reduces noise during low-temperature conditions, and enhances the lifespan of electronic components.

Additionally, the system can be applied in greenhouses and agricultural environments to control ventilation based on temperature levels, supporting healthy plant growth. It can also be integrated into smart home automation systems as an energy-efficient cooling solution. Overall, the automatic temperature-controlled fan system is a versatile and cost-effective solution suitable for various applications where temperature regulation, energy efficiency, and automation are essential.

3.2.5. ADVANTAGES

The automatic temperature-controlled fan project offers several significant advantages over conventional manual fan control systems. One of the major benefits is energy efficiency, as the fan operates only at the required speed based on the surrounding temperature. This prevents unnecessary power consumption when the room temperature is low and reduces electricity wastage, making the system both economical and environmentally friendly.

Another important advantage is enhanced user comfort. Since the fan speed is adjusted automatically, users do not need to manually change speed settings according to temperature variations. This ensures a consistent and comfortable indoor environment, particularly during sleep or long working hours, where frequent manual adjustments are inconvenient. The system also provides fully automated operation, eliminating human intervention. This automation improves reliability and ensures optimal performance even when users are absent.

By avoiding continuous operation at high speeds, the project helps in reducing wear and tear of the fan motor, thereby increasing its lifespan and reducing maintenance costs. The project design is simple, compact, and cost-effective, using easily available components such as Arduino, temperature sensors, and motor drivers. This makes the system easy to implement and suitable for residential, commercial, and small industrial applications. Additionally, the system promotes safe operation by isolating the low-power control circuit from the high-power motor circuit.

Overall, the automatic temperature-controlled fan project demonstrates an efficient, intelligent, and practical solution for modern cooling needs while promoting energy conservation and user convenience. In addition to these benefits, the project supports scalability and flexibility, allowing it to be easily modified or expanded for future requirements. The system can be upgraded with advanced sensors, wireless communication modules, or integration into smart home and IoT platforms without major changes to the existing hardware. Its modular structure makes troubleshooting and maintenance simple, even for beginners.

CHAPTER 4

SOFTWARE AND HARDWARE REQUIREMENTS

4.1. HARDWARE REQUIREMENTS

4.1.1. BREAD BOARD

A breadboard is a reusable prototyping platform used to build and test electronic circuits without soldering. It consists of interconnected metal strips arranged in rows and columns that allow components to be easily inserted and removed. In this project, the breadboard is used to mount and interconnect components such as the DHT11 sensor, jumper wires, and the L298N motor driver. It provides flexibility for modifying connections during testing and debugging stages. The internal power rails of the breadboard help distribute 5V and ground connections efficiently. Using a breadboard reduces the risk of permanent wiring errors and makes the circuit more organized. It is especially useful for educational and experimental projects like this automatic temperature-controlled fan. The breadboard allows rapid prototyping and simplifies troubleshooting. It also helps in verifying circuit operation before implementing a permanent design.

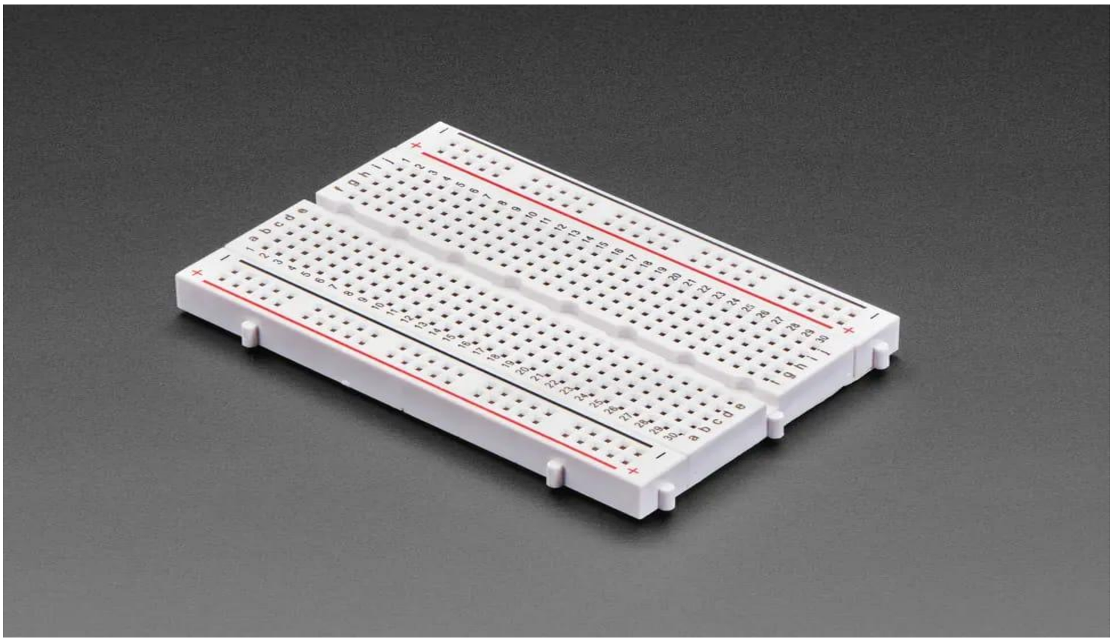


Figure 4.1 Bread Board

4.1.2. ARDUINO UNO

The Arduino Uno is a microcontroller board based on the ATmega328P and serves as the main control unit of the project. It is responsible for reading temperature data from the DHT11 sensor and processing it according to the programmed logic. The Arduino Uno provides multiple digital and analog input/output pins, enabling easy interfacing with sensors and motor drivers. It includes built-in PWM pins that allow precise control of fan speed through the motor driver. The board operates at 5V and can be powered via USB or an external adapter. Its onboard voltage regulator ensures stable operation. The Arduino Uno is widely used due to its simplicity, reliability, and open-source development environment. In this project, it continuously monitors temperature changes and generates appropriate control signals for the fan. The microcontroller eliminates the need for manual fan speed adjustment. Its compact size and low power consumption make it ideal for embedded applications.

Figure 4.2 Arduino uno

4.1.3. 12V DC FAN

The 12V DC fan is the output device of the system and is responsible for providing cooling based on temperature variations. It converts electrical energy into mechanical airflow to reduce ambient temperature. In this project, the fan speed is controlled automatically through the motor driver circuit. DC fans are preferred because their speed can be easily varied using PWM signals. The fan is powered by an external 12V adapter, ensuring sufficient current for stable operation. The fan's speed increases when the temperature rises and decreases when the temperature drops. This controlled operation improves energy efficiency and reduces noise during low-speed operation. The fan is connected to the L298N motor driver, which safely manages power delivery. Using a DC fan ensures safety and compatibility with low-voltage control circuits. The fan demonstrates the practical output of the temperature-controlled system.

Figure 4.3 12V DC Fan

4.1.4. DHT11 TEMP SENSOR

The DHT11 is a digital temperature and humidity sensor used to measure ambient conditions. In this project, it is primarily used for temperature sensing. The sensor contains a calibrated digital output that provides accurate temperature data to the Arduino. It operates on a 5V power supply and consumes very low power. The DHT11 communicates with the Arduino using a single data wire, making it easy to interface. It continuously monitors room temperature and sends updated readings at regular intervals. The sensor's compact size allows easy placement near the fan or control unit. Its low cost and reliability make it suitable for educational projects. The Arduino processes the temperature data from the DHT11 to decide fan speed. The sensor ensures real-time temperature monitoring. It plays a critical role in automating fan operation.

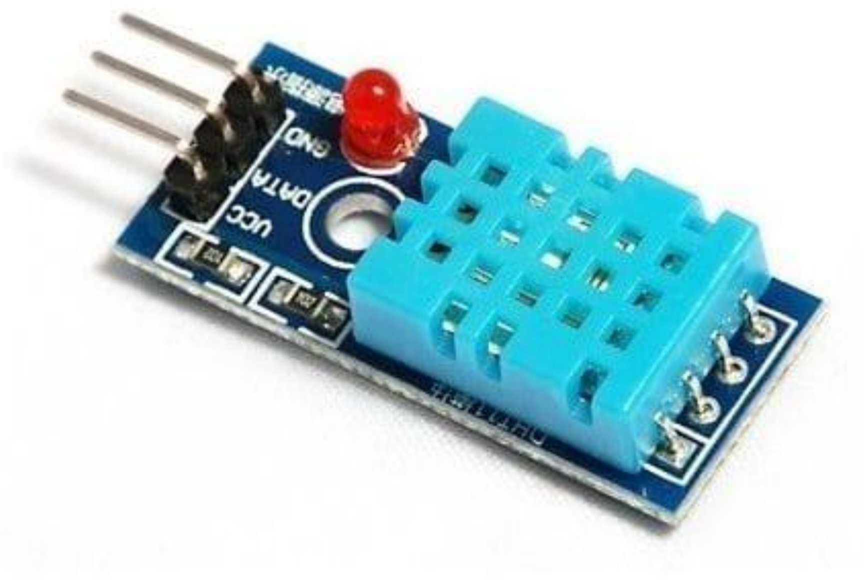


Fig 4.4 DHT11 Temperature Sensor

4.1.5. L298N MOTOR

The L298N motor driver is a dual H-bridge module used to control DC motors safely. In this project, it acts as an interface between the Arduino and the 12V DC fan. Since the Arduino cannot directly supply sufficient current to drive the fan, the L298N amplifies the control signals. It allows speed and direction control of the motor using input signals from the Arduino. The module can handle higher voltages and currents required by the fan. It includes built-in protection diodes to prevent damage from voltage spikes. The L298N receives PWM signals from the Arduino to regulate fan speed. It ensures smooth and efficient motor operation. The module separates the low-power control circuit from the high-power motor circuit. This improves system safety and reliability.

Fig 4.5 L298N Motor

4.1.6. 12V ADAPTER

The 12V adapter provides the required external power supply for the DC fan and the motor driver module. It converts AC mains power into regulated 12V DC output. This ensures stable and continuous operation of the fan. The adapter supplies sufficient current to handle the load requirements of the motor driver and fan. Using an external adapter prevents overloading the Arduino board. The adapter's output is connected directly to the L298N motor driver. A common ground is maintained between the adapter and Arduino for proper signal referencing. The adapter improves overall system performance and reliability. It ensures safe operation by providing isolated power. The use of a 12V adapter is essential for driving high-power components in the system.

Fig 4.6 12V Adapter

4.1.7. JUMPER WIRES

The Jumper wires are used to establish electrical connections between different components on the breadboard and Arduino. They are flexible and available in male-to-male, male-to-female, and female-to-female types. In this project, jumper wires connect the Arduino, sensor, motor driver, and breadboard. They help in transmitting power and control signals throughout the circuit. Jumper wires allow easy modification of connections during testing and debugging. They reduce the need for soldering and speed up circuit assembly. Proper colour coding of jumper wires helps in identifying power, ground, and signal lines. They ensure reliable electrical connections. Jumper wires are essential for building prototype circuits. Their use makes the system neat and organized.

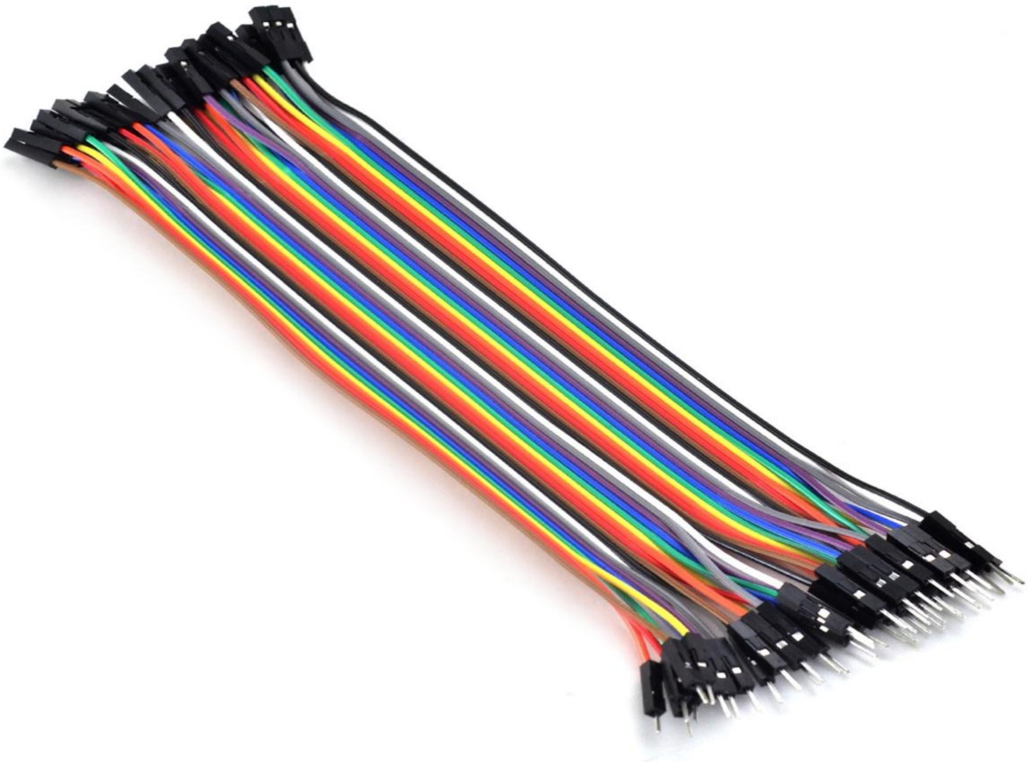


Fig 4.7 Jumper Wires

4.2. SOFTWARE REQUIREMENTS

4.2.1. LAPTOP WITH SOFTWARE (ARDUINO IDE + PROCESSING 3)

A laptop serves as the primary interface for programming and controlling microcontroller-based projects, such as the temperature-controlled fan. It provides the computational platform required to write, compile, and upload code to the Arduino Uno, which is the main controller of this project. The laptop runs Arduino IDE (Integrated Development Environment), which is specially designed to write programs in the Arduino programming language, check for errors, and communicate with the microcontroller via a USB connection. Additionally, Processing 3 can be installed on the laptop to create a graphical interface for visualizing real-time data from the temperature sensor, such as live temperature readings or fan speed levels. This setup allows the user to monitor the performance of the fan, adjust parameters, and debug the system efficiently. The laptop's screen and keyboard provide convenient tools for writing code, editing sketches, and managing libraries required for sensors and motor control. It also allows storage of project files, documentation, and data logs for future analysis. Through the USB port, the laptop supplies a stable connection for serial communication, ensuring smooth data transfer between the Arduino and the software. Overall, the laptop with Arduino IDE and Processing 3 is an essential hardware-software combination that facilitates the design, testing, monitoring, and visualization of the automatic fan system, making the development process faster, more accurate, and user-friendly.

Figure 4.8 Laptop with Software

aic_project_temperature_controlled_fan.ino

```
#include <Adafruit_Sensor.h>
#include <DHT.h>
#include <DHT_U.h>
#include <Wire.h>

#define DHTPIN 7 // DHT11 sensor data pin
#define DHTTYPE DHT11// DHT sensor type
#define MOTOR_PIN_ENA 9 // Enable pin for motor driver (Put pin)
#define MOTOR_PIN_IN1 10 // Input pin 1 for motor driver
#define MOTOR_PIN_IN2 11 // Input pin 2 for motor driver
#define TEMPERATURE_THRESHOLD 29
#define TEMPERATURE_THRESHOLD1 32 // Temperature threshold to adjust motor speed
DHT dht(DHTPIN, DHTTYPE);
void setup() {
  Serial.begin(9600);
  dht.begin();
  Wire.begin();
}
void loop()  {
  delay(2000); // Delay between readings (adjust as needed)
```

Figure 4.9 Coding - 1

aic_project_temperature_controlled_fan.ino

```
  float temperature = dht.readTemperature(); // Read temperature in Celsiu
  if (isnan(temperature)) {
    Serial.println("Failed to read temperature from DHT sensor!");
    return;
  }
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");
  // Adjust motor speed based on temperature
  if (temperature > TEMPERATURE_THRESHOLD1) {
    // Increase motor speed
    analogWrite(MOTOR_PIN_ENA, 255);  // Set the motor speed to maximum (255)
    digitalWrite(MOTOR_PIN_IN1, HIGH);  // Set motor direction (forward)
    digitalWrite(MOTOR_PIN_IN2, LOW);
 }
 else if (temperature > TEMPERATURE_THRESHOLD) {
  // Increase motor speed
   analogWrite(MOTOR_PIN_ENA, 100);   // Set the motor speed to a value (100)
   digitalWrite(MOTOR_PIN_IN1, HIGH);   // Set motor direction (forward)
   digitalWrite(MOTOR_PIN_IN2, LOW);
```

Figure 4.10 Coding - 2

aic_project_temperature_controlled_fan.ino

```
}
else {
 // Decrease motor speed
 analogWrite(MOTOR_PIN_ENA, 45);  // Set the motor speed to a lower value
 digitalWrite(MOTOR_PIN_IN1, HIGH);  // Set motor direction (forward)
 digitalWrite(MOTOR_PIN_IN2, LOW);
  }
 }
```

Figure 4.11 Coding - 3

CHAPTER 5

RESULTS AND DISCUSSIONS

The temperature-controlled fan system was successfully assembled using an Arduino Uno as the central controller, a DHT11 sensor for real-time temperature measurement, and an L298N motor driver to regulate the 12V exhaust fan. During testing, the system consistently detected temperature changes and responded by adjusting the fan operation according to the programmed threshold values. When the temperature exceeded the set limit, the fan turned on immediately and ran at the expected speed, showing effective switching and motor control through the L298N driver. At lower temperatures, the fan remained off, demonstrating reliable hysteresis and preventing unnecessary operation. The use of a 12V adapter and female DC connector ensured stable power delivery, while the breadboard and jumper wires offered flexible prototyping. Overall, the results indicated that the system performed with good stability, responsiveness, and accuracy, proving the feasibility of using Arduino-based automation for temperature-controlled ventilation. Minor variations in temperature readings were observed due to the DHT11's limited precision, but they did not significantly affect system performance. The project demonstrates an efficient and low-cost solution suitable for small-scale environmental control applications.

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Overall, the results show that the system performed efficiently and achieved the intended objective of automatic temperature-based fan control. The response time was quick, and the readings were stable, though minor variations were observed due to the DHT11's limited precision. The cardboard housing helped to organize and protect the components, contributing to a neat and safe prototype. The project demonstrates that an Arduino-based approach is a simple, low-cost, and effective solution for temperature-controlled ventilation in small-scale applications.

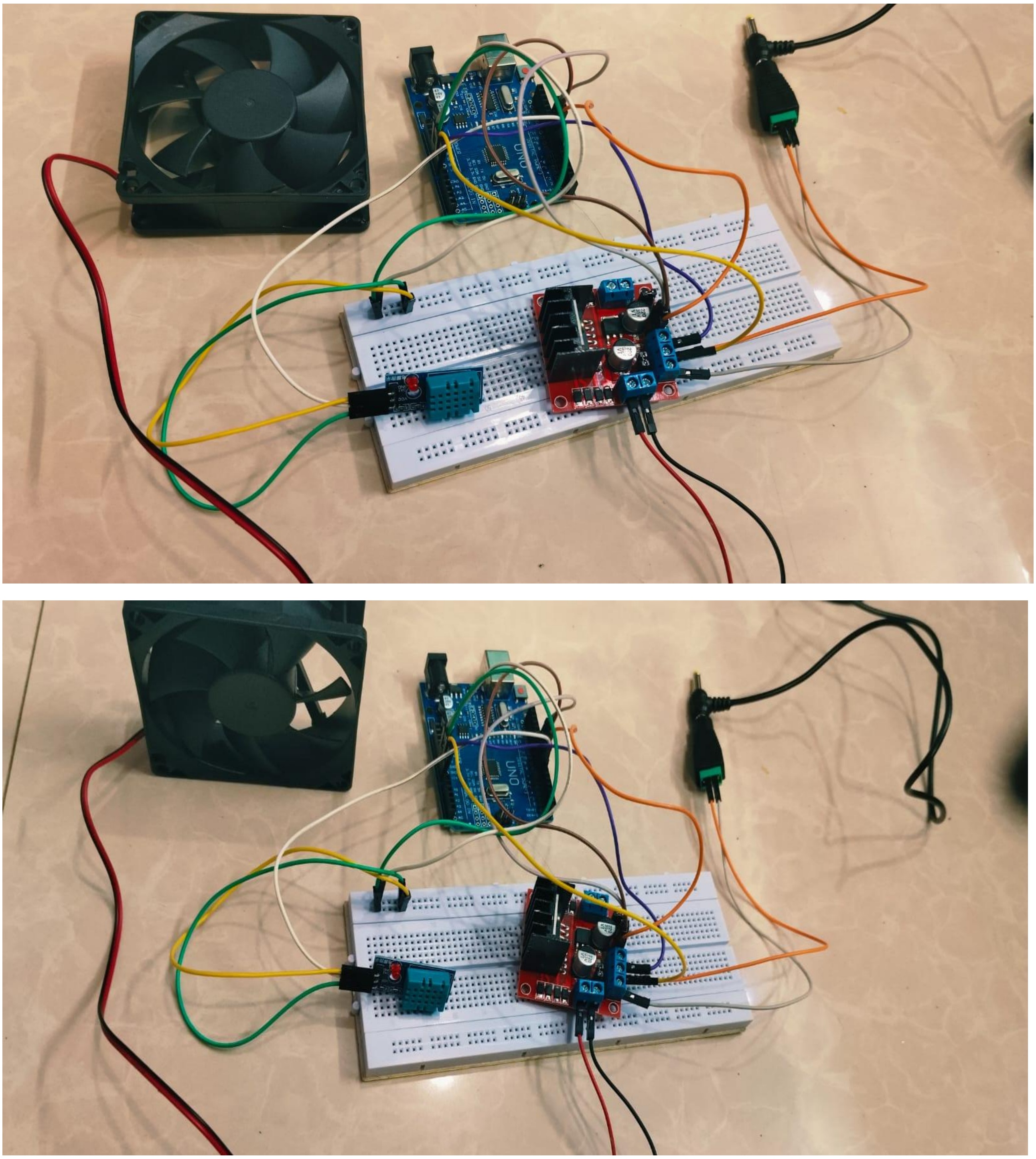

Figure 5.1 Proposed Output Model

CHAPTER 6

CONCULSION

In conclusion, the temperature controller fan system developed using the Arduino Uno, DHT11 sensor, L298N motor driver, and a 12V exhaust fan successfully demonstrates an efficient and low-cost method for monitoring and regulating temperature in a small enclosed space. The DHT11 sensor provided reliable temperature readings, while the Arduino processed the data accurately and controlled the fan operation through the motor driver. The use of a breadboard, jumper wires, and a cardboard enclosure made the setup simple, portable, and easy to assemble for demonstration purposes. Overall, the project achieved its objective by providing a functional temperature-based ventilation system that is easy to build, energy-efficient, and suitable for educational and practical applications.

6.1. FUTURE ENHANCEMENT

One of the possible enhancements is the integration of wireless connectivity, such as Bluetooth or Wi-Fi modules, allowing users to monitor and control the fan speed remotely via a smartphone or computer. Another improvement could be the implementation of automatic temperature thresholds, where the fan speed adjusts more precisely based on finer temperature increments, improving energy efficiency and comfort. The system can also include audible or LED indicators to signal different fan speed levels or temperature ranges, replacing the need for an LCD. Additionally, multiple fans or other cooling devices could be controlled in parallel to maintain a uniform room temperature in larger spaces. Sensors with higher accuracy, like the DHT22, can replace simpler sensors for better performance. Incorporating a humidity sensor alongside the temperature sensor can allow the system to optimize cooling based on both temperature and humidity conditions. The design can also be adapted for industrial or greenhouse applications, where precise climate control is crucial. Finally, the project can be expanded to include energy-saving algorithms, where the fan operates only when necessary, further reducing power consumption. These enhancements make the system more versatile, practical, and adaptable to a wide range of environments.

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